

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.7 / 4.0
- **Coursework:** Data Structures & Algorithms, Linear Algebra for Machine Learning, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

Vylet

May 2026 – Present | vyletdata.com

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Launched Vylet, automating a ~30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Diagnosed a name-collision defect in ownership-verification logic that was incorrectly rejecting valid acquisition targets sharing a name with an unrelated business elsewhere; the fix lifted the pipeline's lead-qualification rate from 79% to 89% with zero change to sourcing volume.

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ~2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ~800 hours of manual pulls across 400 tracked PACs.

PROJECTS

SignalWeaver | *pgvector*, *FastAPI*, *React*, *TypeScript*, *Docker*

github.com/Verdent06/SignalWeaver

Multi-signal financial research platform combining fundamentals, macro indicators, and news sentiment (research assistant; not investment advice)

- Implemented semantic search over stored financial news with *pgvector* cosine similarity (768-d MPNet embeddings), hitting 49ms p50 / 99ms p99 over 90 queries returning top-5 or top-10 matches.
- Built a *React*/*TypeScript* dashboard for composite scores and fundamental/sentiment/macro breakdowns; used it to analyze 90 distinct tickers with scores persisted to Postgres for history views.
- Served composite financial-research scores through async REST endpoints (*FastAPI*) wrapping fundamentals, MPNet sentiment, and regression logic, instrumented end-to-end at 9.1s p50 / 15.2s p99 across 90 successful runs on 90 tickers.
- Containerized the stack with Docker Compose (API + Postgres/*pgvector* + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL, HTML/CSS
Frameworks	React, Angular, FastAPI, Flask, RxJS, LangGraph
Databases	Redis, pgvector
AI / ML	LangSmith, Pandas, ONNX Runtime, Silero VAD
Cloud & Infrastructure	AWS (EC2, S3), Docker, Celery, Git, GitHub Actions